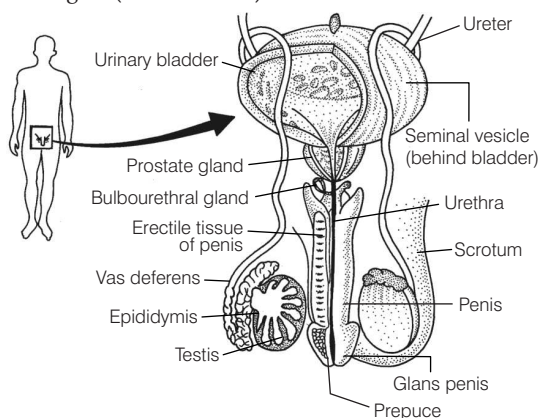


## Human Male Reproductive System

Male reproductive system consists of following main parts:

- **Testes** These are the primary sex organ of males, which perform dual role, i.e. function as endocrine gland and also act as male sex organs. These are suspended in thin skin bag called **scrotal sacs**.
- **Epididymis** It is highly coiled, thin tube formed by union of seminiferous tubules.
- **Penis** It is muscular organ, richly supplied with the blood. Externally, the penis has a bulge called **glans penis**, which is covered by prepuce (a skin fold).
- Male can produce spermatozoa (sperm) throughout their life starting from the age of 13-14 years.
- The growth of hairs on body is due male sex hormones, i.e. androgen (testosterone).



Male reproductive system

### Reproductive Organs, Function and their Numbers in Man

| Reproductive organs of male | Number | Function performed                                                 |
|-----------------------------|--------|--------------------------------------------------------------------|
| Testes                      | 2      | To produce sperms and testosterone.                                |
| Sperm duct                  | 2      | To conduct the sperms from the testes to urethra.                  |
| Seminal vesicles            | 2      | To secrete seminal plasma.                                         |
| Epididymis                  | 2      | To temporarily store sperms and provide mobility.                  |
| Urethra                     | 1      | To conduct urine, sperms.                                          |
| Prostate gland              | 2      | To secrete an alkaline fluid to neutralise the acidity of urethra. |
| Cowper's gland              | 2      | To secrete an alkaline white fluid.                                |
| Penis                       | 1      | To pass urine and deposite sperms in female genital tract.         |

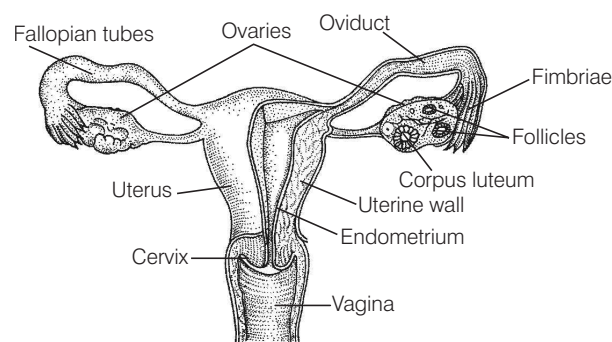
## Human Female Reproductive System

The female reproductive system consists of the following parts:

- **Ovary** These are primary sex organs that produces females gametes (ovum) and several steroid hormone.
- After maturity, the ovary releases an ovum (egg cell) after every 28 days.

### Reproductive Organs, Function and their Number in Female

| Reproductive organ of female | Number | Function performed                     |
|------------------------------|--------|----------------------------------------|
| Ovaries                      | 2      | To produce ova and hormones.           |
| Oviducts                     | 2      | To move the ovum towards uterus.       |
| Uterus                       | 1      | To provide space for developing child. |
| Vagina                       | 1      | To receive the sperms.                 |



Female reproductive system

- **Lactation** Child is delivered after its development and mother produces milk to nourish the child.
- **Placenta** It is the connection between developing embryo and mother. It supplies blood, nutrition, etc., to the foetus.
- **Gestation period** The embryo develops for nine months in uterus. This period is called as **gestation period**.
- **Amniocentesis** A technique to detect chromosomal abnormalities, if any in the developing foetus by analysing the cells in the amniotic fluid. This technique is often misused to know the sex of the foetus. Ultrasound can also be used to detect sex of foetus.

## Menstrual Cycle

- **Reproductive period** of a human female extends from puberty (9-13 years) to menopause (45-50 years).
- **Menopause** is stopping of ovulation and menses. It normally occurs between the ages of 45 and 50. In this stage, women lose the ability to reproduce. Menopause is characterised by hot flushes.
- The periodic vaginal bleeding during menstrual cycle is called **menstruation**.
- Menstrual cycle is controlled by FSH, LH, oestrogen and progesterone.
- On an average, menstrual cycle is completed in 28 days.
- It is absent during pregnancy, may be suppressed during lactation and permanently stops at menopause.
- About 13 mature eggs are released from two ovaries of female in an year.

## Contraception Methods

Contraception is the prevention of union of sperm and ovum.

Some common contraceptive methods are as follow:

- (i) **Natural contraception** Avoid copulation during ovulation.
- (ii) **Mechanical contraception** Use devices to avoid union of sperm and ovum, e.g. condoms, diaphragm, Intra Uterine Devices (IUDs).
- (iii) **Chemical contraception** Fertilisation is avoided by the use of drugs, spermicidal creams, tablets, foam, etc.
- (iv) **Vasectomy** It is the surgical removal of sperm ducts (vas deferens) in males.
- (v) **Tubectomy** Cut and tied of Fallopian tubes in females.

## HUMAN GROWTH

Human population is growing fast and its present rate is about 2% per annum. **Malthus** (1798) gave the concept of **human population**. Study of human population is called **demography**.

- (i) The physiological capacity of organisms is called **biotic potential**.
- (ii) The maximum number of individuals, which an environment can support or sustain is called **carrying capacity**.
- (iii) Population growth is determined by the number of organisms added to the population (through birth) minus the number of organisms lost (through death).
- (iv) When the number of individuals lost is called **zero population**.

Growth curves are of following two types:

- **J-shaped curve** When there is no environmental resistance, a population grows exponentially and a J-shaped curve is obtained.
- **S-shaped curve** When environment resistance does not allow population growth to soar towards infinitely.

# HUMAN HEALTH AND DISEASES

## HEALTH

It is a state of complete physical, mental and social well-being. Any deviation from this state results in a disease.

- According to World Health Organisation (WHO) (1948), 'health is a state of complete physical, mental and social well-being and not merely an absence of disease'.
- There are some basic factors, which contribute to our good health. These are balanced diet, personal hygiene and regular exercise.
- Some other factors having major impact on our health are awareness about diseases and their effects on different functions of body, vaccination against infectious diseases, proper disposal of waste, maintenance of hygienic food and water resources.

## HUMAN DISEASES

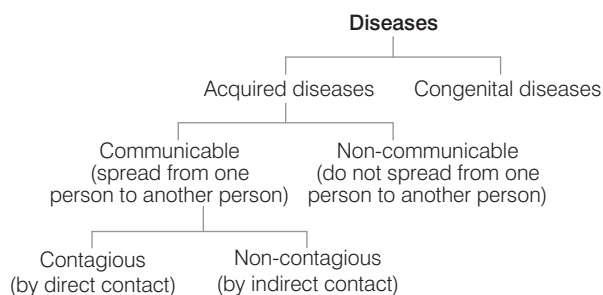
A disease is an abnormal condition affecting the body of an organism or in other words, we can also say that 'any change from normal state causing discomfort is called disease'. It is a specific condition associated with specific symptoms and signs.

- The factors, which affect human health can be categorised into two groups:
  - (i) **Intrinsic factors** The disease causing factors, which are within the human body are called **intrinsic factors**, e.g. malfunctioning of organs (like heart, kidney, etc.) genetic diseases, hormonal imbalance, etc.

- (ii) **Extrinsic factors** The disease causing factors, which come from outside the human body are called **extrinsic factors**, e.g. disease caused by microorganism, tobacco, alcohol, narcotics and inadequate diet.
- The disease causing organisms are called as **pathogens**.
- Various microorganisms responsible for infectious diseases are bacteria, viruses, fungi, protozoans, worms, etc.
- The pathogens enter our body by various ways, multiply and disturb the normal metabolic activities, thus shattering the major organ systems.

## Classification of Diseases

Diseases may be broadly classified into two types:



## Acquired Diseases

These diseases develop after birth and are not transferred from parents to offsprings.

They can be classified into two main types:

- (i) Communicable
- (ii) Non-communicable

## Communicable Diseases or Infectious Diseases

These diseases are caused by pathogens and can spread from an affected person to a normal person. Depending upon the pathogen communicable diseases are of following types:

### (i) Bacterial Diseases

Some of the common bacterial diseases are:

Pneumonia (*Streptococcus*), cholera (*Vibrio cholerae*), tuberculosis (*Mycobacterium tuberculosis*), tetanus (*Clostridium tetani*), leprosy (*Mycobacterium lepreae*), food poisoning (*Salmonella*), plague (*Yersinia pestis*), etc.

- Bacterium *Staphylococcus* present in the nasal passage causes food poisoning by producing a heat resistant toxin in the food.
- Bacterium *Salmonella typhi* is a pathogenic bacterium that causes typhoid fever in human beings. It affects liver, pancreas, nervous system, gastrointestinal tract, kidney, heart, etc.
- Bacterium *Clostridium botulinum* causes food poisoning in canned food.
- Bacteria like *Streptococcus pneumoniae* and *Haemophilus influenzae* are responsible for pneumonia in human beings and infect alveoli of lungs.
- **Anthrax** It is an infectious disease caused by a bacteria called *Bacillus anthracis*. Anthrax cannot directly spread from one individual to other, it spread by its spores.
- **Diphtheria** It is an upper respiratory tract illness caused by *Cornibacterium diphtheriae*, which is characterised by sore throat, low fever and an adherent membrane on the tonsils, pharynx and nasal cavity.
- **Pertusis** (Whooping cough) It is a serious and common infection of children caused by **Gram negative bacteria**, *Bordetella pertusis*. It is an air borne infection with incubation period of 1-3 weeks.
- DPT is a vaccine used against diphtheria, whooping cough and tetanus.
- Conjunctivitis (also called pink eye or madras eye) is inflammation of the conjunctiva (the outermost layer of eye and inner surface of eyelids) caused by adenoviruses. It can also be caused by bacterial infection.
- Rabies is a viral disease that causes acute encephalitis (inflammation of brain). Roughly 97% of human rabies cases come from dog bites, but rabid bats are also capable of causing this disease.
- Hepatitis is a medical condition defined by inflammation of liver. It is caused by hepatitis viruses.
- Influenza commonly called as **flu**, is an infectious disease caused by RNA viruses of family– Orthomyxoviridae.
- Measles also known as rubeola or morbilli is an infection of the respiratory system caused by a virus, specifically a paramyxovirus of the genus–*Morbillivirus*. Symptoms include fever, cough, runny nose, red eyes, etc.
- Mumps is a viral disease of human species caused by mumps virus. Painful swelling of salivary glands (parotid gland) is the most typical symptom.
- **AIDS** (Acquired Immuno Deficiency Syndrome) It weakens the immunity of body. It spreads through sexual contact, transfusion of blood infected by AIDS virus, through infected needles, child gets diseases from mother's milk. Best way to avoid it is to adopt safe sex and use of sterilised syringes.

### Antibiotics

Antibiotics are the chemical substances produced by certain microorganisms that kill or inhibit the growth of other microorganisms. Penicillin was first antibiotic discovered by **Alexander Fleming**. It inhibits cell wall formation in bacteria.

- Antibiotic inhibit the metabolic activities of other microorganisms. Such as synthesis of cell wall protein synthesis, nuclear membrane synthesis, etc.
- The microorganisms, which produce antibiotics (anti-microbial substance) is called **antibiont**.
- The antibiotics can be of bacterial, fungal or Actinomycetes origin.

Some of the antibiotics obtained from different sources are shown below in the given table:

**Microbial Antibiotics**

| Antibiotics  | Source                                                |
|--------------|-------------------------------------------------------|
| Penicillin   | <i>Penicillium notatum</i> ,<br><i>P. chrysogenum</i> |
| Streptomycin | <i>Streptomyces griseus</i>                           |
| Neomycin     | <i>Streptomyces fradiae</i>                           |
| Tetracycline | <i>Streptomyces erythraeus</i>                        |

### (ii) Viral Diseases

Some of the common viral diseases are:

- Polio myelitis (polio) mumps (paramyxovirus), chickenpox (variola virus), measles (paramyxovirus), AIDS (HIV).
- The virus causing dengue fever or break bone fever is an RNA virus of the family– Flaviviridae. Symptoms of dengue include headache, muscle and joint pain.
- Dengue causes platelet count in blood to decrease. It occurs mainly because this virus infects endothelial cells of bone marrow that give rise to platelets resulting in bone marrow suppression.

**(iii) Diseases Caused by Protozoans and Worms**

**Malaria** is caused by *Plasmodium* sp. It is transferred to the host by means of vectors. Insects that can transmit diseases to humans are referred to as vectors, e.g. *Anopheles* serves as a vector for malaria *Culex* for filaria, *Aedes* for dengue and sandfly for kala-azar.

- **Kala-azar** or dum-dum fever is caused by protozoan, *Leishmania donovani*, hence it is also called as **leishmaniasis**.
- The infection to man is caused by the bite of infected sandflies, *Phlebotomus papatasi* and *P. argentipes*.
- **Amoebiasis** It is caused by *Entamoeba histolytica* in man. This protozoan parasitises in large intestine of man, occurring world wide whenever sanitation and hygiene are poor.
- **Sleeping sickness** It is caused by a flagellate protozoan parasite belonging to the class–Mastigophora and genus–*Trypanosoma* and hence, it is also called **trypanosomiasis**. It is transmitted by blood sucking tse-tse fly.
- **Ascariasis** It is a common helminthic infection (worm) caused by *Ascaris lumbricoides*. The disease is characterised by internal bleeding, muscular pain, fever, anaemia and blockage of intestinal passage.
- **Filariasis** It is caused by filarial nematodes, *Wuchereria bancrofti*.

**(iv) Diseases Caused by Fungi**

- Ringworm (skin problems)
- Food poisoning (*Aspergillus*)
- Athlete's foot
- Communicable diseases are of two sub-types on the basis of mode of infection
  - (a) **Contagious Diseases** These diseases are transferred through simple contacts, e.g. smallpox, ringworm, chickenpox, influenza, measles, leprosy, etc.
  - (b) **Non-contagious Diseases** These diseases are transferred through some physical carriers such as air, water, food, etc., e.g. cholera, taeniasis, ascariasis, typhoid, etc. Some of these diseases are transmitted through some **disease carrier vectors**, e.g. plague, AIDS, malaria, hepatitis, etc.

**Non-communicable Diseases or Non-Infectious Diseases**

These diseases remain confined only to the person who develops them. They can be of four types:

- (i) **Organic** (degenerative) **diseases** caused due to organ malfunctioning.
- (ii) **Deficiency diseases** caused due to deficiency of mineral, hormones, vitamins and nutrients.
- (iii) **Allergic diseases** caused due to the hypersensitivity of body to foreign particle.
- (iv) **Cancer** caused due to uncontrolled division of certain cells of body.

**Diseases of Malfunctioning**

Diseases caused by malfunctioning of organs are:

- Disease caused by improper functioning of heart–**Cardiac failure**.
- Disease caused by improper functioning of kidney–**Kidney failure**.
- Disease caused by improper functioning of bones–**Osteoporosis**.
- Disease caused by improper functioning of eyes–**Myopia, hypermetropia and cataract**.

**Deficiency Diseases**

The diseases caused due to defect in the production of hormones are deficiency diseases. These are as follows:

- **Cretinism** In children, occurs due to deficiency of thyroxine hormone, which makes a child mentally retarded.
- **Diabetes** Due to deficiency of insulin.
- **Dwarfism** Due to the sluggish activity of the pituitary gland, the person has diminished growth of the bones.
- **Gigantism** Due to the excessive activity of the pituitary gland, the person has excessive growth of the bones.
- **Goitre** It is caused by deficiency of iodine in diet. It is characterised by swelling of thyroid gland.
- **Anaemia** It is a decrease in number of RBCs or less than the normal quantity of haemoglobin in the blood. Anaemia leads to hypoxia (lack of oxygen) in organs.

**Allergic Diseases**

The unfavourable response of the body to certain things like dust, serum, drugs, fabrics and pollen, etc., is called **allergy**. Sneezing, irritation of throat, itching and skin rashes, etc., are the symptoms of allergic diseases, e.g. asthma is caused by allergy.

**Cancer**

It is an uncontrolled growth of normal cells, which have lost control on cell division as a result these cells spread in the body. Sometimes cells keep dividing and produce more cells even when they are not required. As a result, mass of tissue is formed, which is called as **tumour**. Tumours can be benign or malignant.

**Types of Cancers**

- (a) Carcinomas, e.g. lung cancer, breast cancer, cancer of pancreas and stomach.
- (b) Sarcomas, e.g. bone cancer.
- (c) Leukemia, e.g. blood cancer.

**Breast cancer** It is common among women while lung cancer is common among smokers.

- Radiations cause cancer, leukemia and skin injuries damage genes or chromosomes, induce unmarked cell division, destroy tissues, cells and blood cells.
- Cancer is common in older people because over the time their genomes tend to accumulate more mutations.



### Diseases Caused by Addictive Substances

**Alcohol** Reduces alertness of mind and causes liver damage.

**Narcotic drugs** Narcotic drugs are morphine, cocaine, heroin, hashish, ganja, opium, marijuana, barbiturates and LAD (Lysergic Acid Diethylamide).

### Diseases Caused due to Pollution

**Chemical pollution** Pneumoconiosis is caused by inhaling coal dust. Silicosis is caused by inhaling stone dust (silica). Asbestosis is caused by inhaling asbestos dust. Lung diseases and bronchitis are caused by smoke.



#### IMPORTANT POINTS TO REMEMBER

DOTS or directly observed treatment, short-course was launched by WHO as a strategy to control TB (Tuberculosis). It ensures a regular, uninterrupted supply of high quality anti-TB drug.

### Congenital Diseases

These are anatomical or physiological abnormalities that come from birth. These diseases can be caused by genetic mutations or environmental factors. Diseases by genetic mutation can be transmitted to next generation, whereas diseases by environmental factors are non-transmissible.

Some common congenital diseases are phenylketonuria, alkaptonuria, haemophilia, sickle-cell anaemia, etc.

### Chromosomal Aberrations

These are any variation from the 46 chromosome number of humans. These include Turner's syndrome, Klinefelter syndrome, etc.

## VACCINATION

**Louis Pasteur** coined the term 'Vaccine' and vaccination for the first time.

- The word 'vaccination' has been derived from a Latin word '*vacca*' which means **cow**.
- A vaccine is either a cell suspension or a byproduct excreted by the cell.
- These vaccines, when introduced into the body, stimulate the production of antibodies.
- Serum is used for obtaining antibodies.
- This introduction of harmless antigenic material into healthy person for providing acquired immunity against the disease is known as **immunisation** and **vaccination**.

#### Diseases of Animals

| Diseases        | Animal affected     | Caused by | Organ affected         |
|-----------------|---------------------|-----------|------------------------|
| Pox             | Cattle, sheep, goat | Virus     | Skin                   |
| Tuberculosis    | Cattle, birds       | Bacteria  | Lungs, intestine       |
| Rinderpest      | Cattle, sheep, goat | Bacteria  | Spleen                 |
| Anthrax         | Cattle, sheep, goat | Bacteria  | Skin, lungs, intestine |
| Foot and mouth  | Cattle, pigs        | Virus     | Pharynx                |
| Dermatitis      | Goat, sheep, cattle | Virus     | Skin                   |
| Mastitis        | Cattle              | Bacteria  | Swollen udders         |
| Ringworm        | Cattle              | Fungus    | Body                   |
| Trypanosomiasis | Cattle              | Protozoan | Body                   |
| Helminthic      | Cattle              | Worms     | Body                   |
| Aspergillosis   | Poultry birds       | Fungus    | Body                   |
| Cholera         | Poultry birds       | Bacteria  | Body                   |
| Diarrhoea       | Poultry birds       | Bacteria  | Body                   |
| Ranikhet        | Poultry birds       | Virus     | Body                   |

## APPLIED BIOLOGY

### FERTILISERS

Fertiliser is any material found naturally or formed synthetically and is applied to soil or plant tissues, to supply one or more plant nutrients essential to the growth of plants.

They are of mainly two types:

#### 1. Chemical Fertilisers

These are salt or an organic compound containing the necessary plant nutrients. They may be organic or inorganic, e.g. ammonium sulphate is an inorganic fertiliser, while urea is an organic fertiliser. If excess fertilisers are applied to a plant without water, crop will die due to the plasmolysis.

Chemical fertilisers can be classified into following types:

- Nitrogenous fertilisers** These are nitrogen containing fertilisers, e.g.
  - Ammonium nitrate  $[\text{NH}_4\text{NO}_3]$
  - Ammonium sulphate  $[(\text{NH}_4)_2\text{SO}_4]$
  - Sodium nitrate  $[\text{NaNO}_3]$
  - Urea  $[\text{NH}_2\text{CONH}_2]$  (organic fertiliser)
- Phosphatic fertilisers** These are phosphorus containing fertilisers, e.g.
  - Ammonium phosphate  $[(\text{NH}_4)_3\text{PO}_4]$
  - Ammonium hydrogen phosphate  $[(\text{NH}_4)_2\text{H}_2\text{PO}_4]$
  - Superphosphate  $[\text{Ca}_2(\text{H}_2\text{PO}_4)_2]$
- Potassium fertilisers** These are potassium containing fertilisers, e.g.
  - Potassium chloride  $[\text{KCl}]$
  - Potassium nitrate  $[\text{KNO}_3]$